When working on an electronics project, various electrical parameters such as voltage and current need to be considered for proper implementation and functioning of your project. This chapter touches upon the necessary parameters that a beginner should necessarily be aware of. We'll then look at some ways to troubleshoot your project and finally learn how to use the power supply of your desktop computer to power your Arduino™ hardware.

Parameters of importance

- **Voltage:** Voltage is actually a difference in electrical energy (called 'potential') between two points. So when we say "connect the device to a voltage of 5V (5 volts)", you can connect the device between 5V and GROUND (0V) which is 5-0=5V, or you can connect between 10V and 5V giving 10-5=5V.
- **Current:** When voltage that's more than zero is applied, electrons carry energy, and the flow of these electrons is called 'current'. This is measured in Amperes (A).
- **Power (denoted by P):** The power a device uses, generally measured in watts, is the product of voltage and current.

The USB port 3.0 of our computers provide 5V voltage and current of 1A. So the maximum power it can supply is P=V x I = 5 x 1= 5W. So when we charge a mobile phone or an MP3 player via our computers, 5 watts is the maximum rate of charging.

Power is a parameter that's usually ignored by beginners in electronic design. However, it needs to be taken into consideration when connecting devices. Connecting high power devices directly can damage circuits and devices they're connected to. So check for power requirement when you connect your next device. Devices such as motors need lot of power for their operation and hence should never be connected directly in the circuit. They need special chips called motor drivers, which can supply the required amount of power.

Terminology

- **GROUND:** It's assumed that the GROUND is always 0V. When used in a circuit, it's denoted by the symbol ⏚. Remember '0V' and 'GROUND' are both used interchangeably.